

(GPRS), a module that would allow the user to access the internet while on the move. This culminated in the 1999 update which led to the inclusion of enhanced data rates for GSM evolution (EDGE) in the GSM standard. EDGE allowed for higher bandwidth internet connectivity.

GSM is a cellular telephone standard. This means that base stations with which the mobile phones communicate form cells. The mobile phone communicates by searching for the closest cell with the highest signal strength. While the phone is on

GSM has its origins in Europe with nations coming together to unify their wireless communication standards in the early 70s

the move, the mobile phone has to switch from one cell to another. This process is called handing over.

As a second generation mobile telecommunication standard, the challenge faced by GSM was that it had to allow multiple cell phones to use the same frequency spectrum. It performed this through a technique known as time division multiple access (TDMA). Under this scheme,

the conversation of one user for a small finite duration is recorded, encoded and transmitted in a burst that lasts much shorter than the duration of the conversation. Thus, at any given moment of time, only one of the many cell phones within a cell is transmitting at the given frequency. This procedure is known as time division multiplexing. The maximum number of simultaneous calls a cell can handle is given by the length of the finite duration of conversation divided by the length of the burst. This varies depending on factors such as the type of cell and the standard used.

GSM uses a specific set of frequencies. In India, all providers use either the 900 MHz or the 1,800 MHz spectrum. Most European networks operate at 2,100 MHz, while Canadian and American networks utilise the 950 MHz spectrum. Each of these frequencies is known as a band of the GSM spectrum. A phone handset is said to be Quad Band if it can operate at all these frequencies. Similarly, Tri Band if it can operate in three out of the four frequency bands.

In GSM, each connection is identified by a unique SIM card, while each handset is identified by an international mobile